

# Cybersecurity Landscape & Careers

What cybersecurity is, who defends against threats, and how to build a career in the field — with hands-on labs and applied exercises.

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Duration: 60 minutes | Level: Beginner

# Agenda

**01**

## **What Is Cybersecurity?**

Core concepts, why it matters, and today's threat landscape

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**02**

## **Roles & Responsibilities**

Who does what on a security team, and how they collaborate

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**03**

## **Careers in Cybersecurity**

Career pathways, in-demand skills, and how to get started

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*Each topic includes a hands-on lab and an applied exercise to build real-world readiness.*

● TOPIC 1 OF 3

# What Is Cybersecurity?

Core concepts, motives, and the shape of today's threat landscape

## KEY TERM

# Cybersecurity

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The practice of protecting systems, networks, applications, and data from digital attacks, unauthorized access, damage, or disruption — across every layer of an organization.

*It spans three goals, known as the CIA Triad — Confidentiality, Integrity, and Availability — covered later in this module.*

# \$10.5T

projected annual global  
cost of cybercrime by 2025

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Source: Cybersecurity Ventures

# Why it matters right now

The scale of the problem, in three numbers

**11 sec**

Estimated interval  
between ransomware  
attacks worldwide

**\$4.45M**

Average cost of a  
data breach globally

**3.5M+**

Unfilled cybersecurity  
jobs worldwide

*Figures are commonly cited industry estimates (Cybersecurity Ventures, IBM, ISC2)*

# The expanding attack surface

Every new connected system is a new door to defend

## • Cloud & remote work

Employees and data now live outside the traditional office perimeter, across SaaS apps and home networks.

## • IoT & connected devices

Cameras, sensors, and smart devices often ship with weak default security and rarely get patched.

## • Mobile devices

Phones and tablets carry corporate data and credentials, often on personal, less-controlled networks.

## • Supply chain & third parties

Vendors and contractors with system access can become the weakest link in an otherwise secure org.

## REAL-WORLD CASE STUDY

*“In 2013, attackers accessed a major retailer's payment systems using credentials stolen from a small HVAC vendor — exposing 40 million customer payment cards during the holiday season.”*

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### What went wrong

A third-party vendor had far more network access than its job required, and that access wasn't segmented from payment systems.

### Takeaway

Security is only as strong as the weakest connected party.  
Vendor access and network segmentation are core defenses, not afterthoughts.

# Explore a live threat intelligence map

Objective: get comfortable navigating real-time global threat data

## ENVIRONMENT

Web browser only — no installation required

- 1 Open a live threat map tool (e.g., a major vendor's public cyberthreat map)
- 2 Identify the top 3 attack types currently active worldwide
- 3 Note which countries or regions show the highest attack volume right now
- 4 Cross-reference one attack type you see with a term from this module

**Expected outcome: A screenshot plus a 3-sentence summary of the patterns you observed**



## APPLICATION EXERCISE

# Analyze a real breach

Apply what you've learned to a real incident — individually or in pairs

### SCENARIO

Choose a well-documented breach (for example: a major retailer, a critical infrastructure operator, or a credit bureau) and research it using public reporting.

### YOUR TASK

- ✓ Identify the attack vector — how did attackers get in?
- ✓ Estimate the business impact — cost, downtime, reputational damage
- ✓ Propose one control that could have prevented or limited it

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*Deliverable: A 1-page breach analysis brief | Time: 30 minutes*

## Which best describes the primary goal of cybersecurity?

**A** Preventing all employees from accessing the internet

**B** Protecting systems, networks, and data from unauthorized access or damage

✓ **Correct**

**C** Only stopping viruses from infecting computers

**D** Making sure software updates are installed monthly

● TOPIC 2 OF 3

# Roles & Responsibilities

Who does what on a security team, and how they work together

# Core security roles

The people behind the systems that keep an organization safe

## ● SOC Analyst

Monitors alerts and investigates suspicious activity around the clock.

## ● Security Engineer

Builds and maintains the tools and infrastructure that enforce security controls.

## ● Penetration Tester

Simulates attacks to find weaknesses before real attackers do.

## ● Incident Responder

Leads the response when a breach or security event is confirmed.

## ● Security Architect

Designs how systems and networks are structured to resist attack.

## ● CISO

Owns security strategy and reports risk to executive leadership.

# Tactical vs. strategic responsibility

Every security team needs both hands-on defenders and big-picture owners

## Tactical (SOC Analyst)

- Triage alerts as they come in, in real time
- Follows documented playbooks to contain known threats
- Escalates anything unfamiliar to a senior analyst

## Strategic (CISO)

- Sets security policy and long-term investment priorities
- Reports risk posture and incidents to the board
- Owns compliance with standards like ISO 27001 or NIST

## HANDS-ON LAB

# Map roles to a security incident

Objective: understand how roles collaborate when something goes wrong

**ENVIRONMENT**

Provided incident scenario handout (no software required)

- 1 Read the provided incident scenario (a suspected ransomware alert)
- 2 Identify which role responds first, and what they do in the first 15 minutes
- 3 Map the escalation path from first responder up to leadership
- 4 List which role is responsible for communicating with affected customers

**Expected outcome: A completed roles-to-incident flow diagram**

## APPLICATION EXERCISE

# Build a responsibility matrix

Turn today's roles into a practical, reusable planning tool

### SCENARIO

A phishing email led to a compromised employee account. Five roles are involved in the response.

### YOUR TASK

- ✓ List the 5 roles: SOC Analyst, Incident Responder, Security Engineer, IT Helpdesk, CISO
- ✓ For each, mark Responsible, Accountable, Consulted, or Informed (RACI)
- ✓ Flag any gap where no one is clearly Responsible

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*Deliverable: a completed RACI matrix | Time: 20 minutes*

## Who typically owns security strategy and board-level risk reporting?

**A** SOC Analyst

**B** Penetration Tester

**C** CISO

✓ Correct

**D** IT Helpdesk



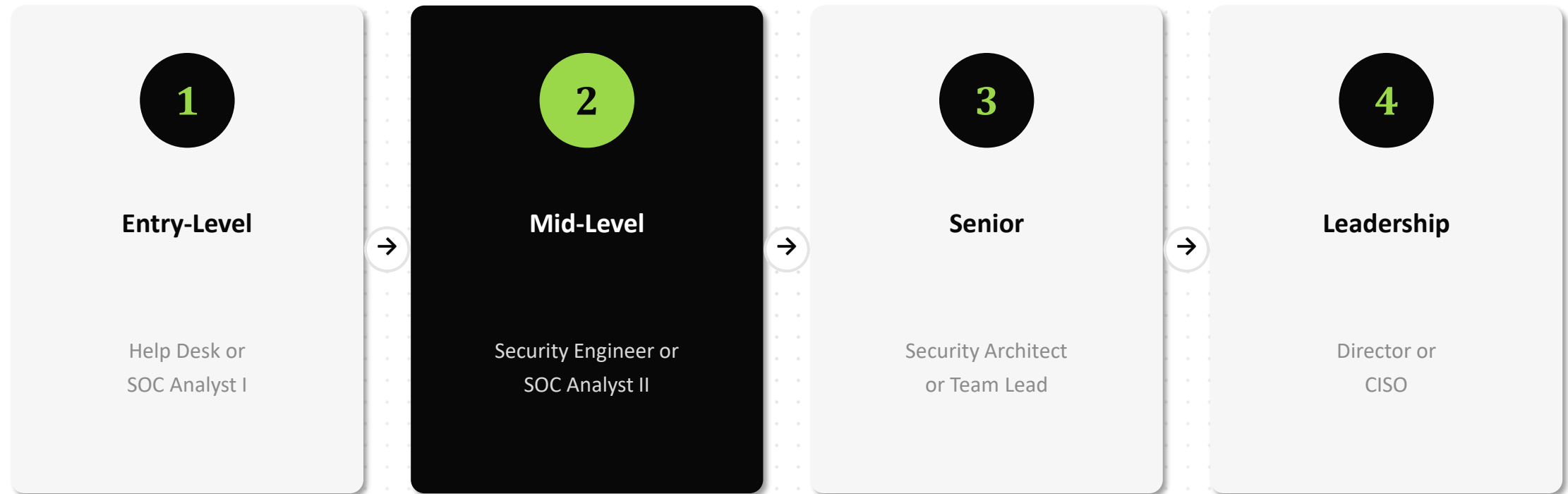
● TOPIC 3 OF 3

# Careers in Cybersecurity

Career pathways, in-demand skills, and how to get started

# A typical career pathway

Most cybersecurity careers follow a recognizable progression



# In-demand skills & certifications

What hiring managers look for when screening entry-level candidates

## Technical Skills

- ✓ Networking fundamentals (TCP/IP, DNS)
- ✓ Basic scripting (Python or Bash)
- ✓ Cloud platform basics (AWS/Azure)

## Certifications

- ✓ CompTIA Security+
- ✓ Certified Ethical Hacker (CEH)
- ✓ CISSP (for senior roles)

## Soft Skills

- ✓ Clear incident documentation
- ✓ Cross-team communication
- ✓ Staying calm under pressure

# Job market & demand

Cybersecurity remains one of the most resilient tech job markets

**32%**

Projected job growth  
for security roles  
through 2032

**0%**

Unemployment rate  
in cybersecurity  
(effectively full employment)

**\$95K+**

Typical starting salary  
for entry-level analyst  
roles in the US

*Figures are commonly cited industry estimates (U.S. BLS, ISC2 Workforce Study)*

# Build your cybersecurity career roadmap

Objective: turn today's content into a personal 12-month plan

## ENVIRONMENT

Provided roadmap worksheet template (digital or printed)

- 1 Assess your current skills against the entry-level checklist from this module
- 2 Pick one target entry role (e.g., SOC Analyst I)
- 3 List 3 certifications or courses you'll need to qualify
- 4 Set 3 milestones with target dates over the next 12 months

**Expected outcome: a completed, personalized 12-month roadmap worksheet**

## APPLICATION EXERCISE

# Resume & LinkedIn positioning

Translate what you've learned into how you present yourself to employers

### SCENARIO

Recruiters scan resumes and LinkedIn profiles for specific keywords tied to the role they're hiring for.

### YOUR TASK

- ✓ Rewrite one resume bullet to highlight a security-relevant skill or project
- ✓ Draft a LinkedIn headline targeting your chosen entry-level role
- ✓ Add one certification or course from today's list as "in progress"

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*Deliverable: A before/after comparison of one resume bullet and one headline | Time: 20 minutes*

## Which certification is most commonly recommended for entry-level roles?

**A**

CompTIA Security+

✓ Correct

**B**

CISSP

**C**

A four-year computer science degree only

**D**

No certification is ever useful

# Key takeaways

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- 1 Cybersecurity protects systems, networks, and data as the attack surface keeps expanding.
- 2 Security teams need both tactical defenders (SOC analysts) and strategic owners (CISOs).
- 3 Career paths are well-defined: entry-level roles lead to engineering, architecture, and leadership.
- 4 Certifications like Security+ and hands-on practice matter as much as formal education.



# References & resources

Sources cited in this module, and where to go deeper

## Standards & Frameworks

- NIST Cybersecurity Framework (CSF)
- ISO/IEC 27001
- MITRE ATT&CK Framework

## Reports Cited

- Cybersecurity Ventures — Cybercrime Report
- IBM Cost of a Data Breach Report
- ISC2 Cybersecurity Workforce Study

## Communities & Further Reading

- Krebs on Security (blog)
- r/cybersecurity
- Local OWASP chapter meetups

*Statistics in this module are commonly cited industry figures.*



# Section Complete

Next: Module 01 — Domains of Cybersecurity & Cloud Security

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Questions? [security-training@bsoft.com](mailto:security-training@bsoft.com)